

Antoine Belley | Curriculum Vitae

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Postdoctoral Fellow of the [@NDB NSF Hub](#), located at the Institute for Nuclear Theory at the University of Washington.

Work Experience

- **Postdoctoral Fellow of the @NDB NSF HUB** **Institute For Nuclear Theory, UW, USA**
Supervisor: Prof. Vincenzo Cirigliano Oct. 2026–
- **Postdoctoral Fellow** **MIT, USA**
Supervisor: Prof. Ronald Fernando Garcia Ruiz Oct. 2024–Sept-2026

Education

- **Doctor of Philosophy in Physics** **Canada**
University of British Columbia (UBC)/TRIUMF, Supervisor: Dr. Jason Holt 2020-2024
- **Bachelor of Science** **Canada**
McGill University 2017-2020
First Class Honours in Mathematics and Physics with Distinction

Publication summary

As of September 29, 2026:

- **389 citations** on Google Scholar, h-index = 8, i10-index = 7
- Principal author in high impact publications:
 - **Physical Review Letters** (3 published, 1 under review)
 - **Nature** (1 under review)
 - **Review of Modern Physics** (1 under review)
- **16 publications:**
 - 8 peer reviewed
 - 6 under review
 - 2 conference proceedings
 - **First author in 8 of these publications**

Community service

- **Referee for Nuclear Physics A, Physics Letter B, Physical Review X, Physical Review Letters and Nature**
- **Organizer** **ECT*, Italy**
Workshop on “Synergy between nuclear theory and quantum sensing experiments for fundamental physics” 2027
- **Organizer** **ESNT, France**
Workshop on “Theoretical and experimental developments for symmetry-violating nuclear properties” Jun. 2025
- **VP Internal** **McGill University, CA**
McGill Society of Physics Students Sept. 2019 - May 2020
- **VP Finance** **McGill University, CA**
Canadian Undergraduate Physics Conference 2019 (CUPC 2019) Oct. 2018 - Dec. 2019

Supervision Experience

- Co-supervised 5 graduate students
- Co-supervised 15 undergraduate students

Teaching Experience

- **Teacher Assistant/Guest lecturer** **TRIUMF, CA**
UBC Phys505:Nuclear Physics Jan. 2023 - Apr. 2023

Grants and Funding

- **MGAIC Innovative Project** **MIT**
\$100,000, Towards a Foundation Model for Atomic Nuclei 2026-2027
Co-applicant, PI: Ronald Fernando Garcia Ruiz

Scholarships and Fellowships

- **NSERC Postdoctoral Fellowship [PDF]** **MIT**
\$70,000/year 2024-2026
- **NSERC Alexander Graham Bell Canada Graduate Scholarships-Doctoral [CGSD]** **UBC**
\$35,000/year 2021-2024
- **Four Year Doctoral Fellowship** **UBC**
Accepted in name only 2021-2025
- **Fonds de recherche du Québec - Bourses de maîtrise en recherche** **UBC**
\$17,500/year, declined due to better funding 2021-2022
- **NSERC Alexander Graham Bell Canada Graduate Scholarships-Master's [CGSM]** **UBC**
\$17,500/year 2020-2021
- **Undergraduate Research Scholarship - Canadian Institute of Nuclear Physics** **TRIUMF**
\$4000 2019
- **Fonds de recherche du Québec- Suppléments à la bourse BPC du CRSNG** **McGill University**
\$2000 2018
- **NSERC Undergraduate Student Research Awards [USRA]** **McGill University**
\$4500 2018

Awards and Distinctions

- **Carl H. Westcott Fellowship** **TRIUMF**
\$11,000 2023
Yearly award given to one student affiliated to TRIUMF
- **Faculty of Science PhD Tuition Award** **UBC**
Tuition 2021-2025
- **President's Academic Excellence Initiative PhD Award** **UBC**
\$1300/year 2021-2025
- **Best Presentation in Theoretical Physics** **Winter Nuclear & Particle Physics Conference, Virtual**
\$500 2021
- **First place - Physics Department Summer Student Poster Presentation** **McGill University**
\$100 2019

- **First prize in physical science** **McGill Undergraduate Research Conference**
\$300 2018
 - **Second place - Physics Department Summer Student Poster Presentation** **McGill University**
\$50 2018
 - **Travel Grant to CUPC 2018** **Canadian Institute of Nuclear Physics**
\$500 2018
 - **Bourse persévérance et rayonnement** **Collège Bois-de-Boulogne**
\$500 2017
 - **Prix Carolyne-Dion** **Département de physique, Collège Bois-de-Boulogne**
\$100 2017
- Awards chosen by the students for the volunteer help given to his peers

Presentations

- Invited Seminars and Colloquia**
- **T-2 Seminars** **Los Alamos National Lab**
Using the atomic nucleus as a probe for BSM physics. Jun. 2 2026
 - **Sun Yat-sen University Physics Seminars** **Virtual**
Applications of machine learning emulators in nuclear physics. May 6 2026
 - **Oak Ridge National Lab Physics Seminars** **Virtual**
Using the atomic nucleus as a probe for BSM physics. Feb. 4 2026
 - **Institute of Theoretical Physics Seminars** **Ruhr-Universität Bochum, DE**
Applications of machine learning emulators in nuclear physics. Jan. 29 2026
 - **Neutrinoless Double Beta Decay Hub @NDB Seminars** **Virtual**
Ab initio theory towards reliable neutrinoless double beta decay nuclear matrix elements. Dec. 5 2025
 - **Saint-Mary's University Physics Colloquium** **Virtual**
Using the atomic nucleus as a probe for BSM physics. May 26 2025
 - **Institute for Nuclear Theory - Rising Researcher Seminar** **Virtual**
Using the atomic nucleus as a probe for BSM physics. Feb. 11 2025
 - **Nuclear Physics Seminar** **University of Notre Dame, USA**
Using the atomic nucleus as a probe for BSM physics. Nov. 18 2024
 - **FRIB Seminar** **FRIB, MSU, USA**
Ab initio theory towards reliable neutrinoless double beta decay nuclear matrix elements. Dec. 7 2023
 - **HEP/Astro Results Forum** **Virtual**
Status of neutrinoless double-beta decay from ab initio nuclear theory. Sept. 28 2023
 - **Institut Für Kernphysik Seminars** **TU Darmstadt, DE**
The road towards reliable nuclear matrix elements for neutrinoless double beta decay. Jul. 5 2023
 - **Particle and Astroparticle Physics Seminars** **McGill University, CA**
The road towards reliable nuclear matrix elements for neutrinoless double beta decay. Sept. 19 2022

- Invited Conference Presentations**
- **NXT-AI-2026** **Boston, USA**
Building a foundation model of the atomic nucleus Nov. 3 2026
 - **IMSRG Flow and Tell meeting** **University of Notre Dame, USA**
Building a foundation model of the atomic nucleus Oct. 4 2026

- **Nuclear Structure 2026** Simon Fraser University, CA
The Standard Model and Beyond from a Nuclear Structure Perspective. Jul. 27 2026
Only postdoctoral researcher to be invited to speak.
- **INT PROGRAM INT-26-2a: Recommended Values for $0\nu\beta\beta$ Nuclear Matrix Elements** INT, USA
Uncertainty quantification of $0\nu\beta\beta$ NMEs Jul. 17 2026
- **XeSAT2026** TRIUMF, CA
Ab initio theory towards reliable neutrinoless double beta decay nuclear matrix elements. May 19 2026
- **INT PROGRAM INT-26-1: Nuclear Hamiltonians for Advancing Nuclear Physics and Beyond** INT, USA
Connecting nuclear force parameters to nuclear observables via machine learning emulation Apr. 28 2026
- **Progress on ab initio nuclear theory** TRIUMF, CA
Violating Parity in the In Medium Similarity Renormalization Group Feb. 24 2026
- **The search for double- β decay: experimental and theoretical challenges** ECT*, IT
Ab initio theory towards reliable neutrinoless double beta decay nuclear matrix elements. Feb. 3 2026
Keynote speaker.
- **Quantum Science and Artificial Intelligence**
for Fundamental Physics Universidad Nacional de Colombia, CO
Global emulation of atomic nuclei. Dec. 18 2025
- **IMSRG Flow and Tell Meeting** University of Notre Dame, USA
Global emulation of atomic nuclei in the IMSRG. Oct. 21 2025
- **Standard Model EFT meets Chiral EFT** TRIUMF, CA
Violating Parity in the In Medium Similarity Renormalization Group Oct. 3 2025
- **Theoretical and experimental developments for symmetry-violating nuclear properties** ESNT, FR
Parity Violating In Medium Similarity Renormalization Group Jun. 26 2025
- **Conference at the Intersection of Particle and Nuclear Physics** University of Wisconsin-Madison, USA
Ab initio neutrinoless double beta decay matrix elements Jun. 12 2025
- **Progress on ab initio nuclear theory** TRIUMF, CA
Attention everyone! I have gone completely BANNANE! Feb. 28 2025
- **NME 2025: Theoretical and experimental approaches for NME of double-beta decay** RCNP, JPN
Using the atomic nucleus as a probe for BSM physics. Jan. 21 2025
- **Fall Meeting of the APS Division of Nuclear Physics** MIT, USA
The road towards reliable neutrinoless double-beta decay NMEs. Oct. 7 2024
- **Celebrating 75 Years of the Nuclear Shell Model and Maria Goeppert-Mayer** Argonne National Lab, USA
Ab initio theory towards reliable $0\nu\beta\beta$ nuclear matrix elements. Jul. 21 2024
- **Progress on ab initio nuclear theory** TRIUMF, CA
Ab initio theory towards reliable $0\nu\beta\beta$ nuclear matrix elements. Feb. 28 2024
- **INT PROGRAM INT-23-1B: New physics searches at the precision frontier** INT, USA
Towards reliable nuclear matrix elements for neutrinoless double beta decay. May 22 2023
- **Progress on ab initio nuclear theory** TRIUMF, CA
The road towards reliable nuclear matrix elements for neutrinoless double beta decay. Mar. 2 2023
- **2nd Joint Canada-APCTP Meeting** TRIUMF, CA
Constraining $0\nu\beta\beta$ decay matrix elements from ab initio nuclear theory. Aug. 10 2022
- **MEDEX'22** CTU, Prague, CZE
Constraining $0\nu\beta\beta$ decay matrix elements from ab initio nuclear theory. Jun. 16 2022

Contributed Conference Presentations.....

- **Fall Meeting of the APS Division of Nuclear Physics** **Chicago, USA**
Talk on "Global emulation of atomic nuclei." Oct. 19 2025
- **IAIFI Summer Workshop** **MIT, USA**
Talk on "Emulating the atomic nucleus." Aug. 10 2025
- **Workshop on Basic Computing Services in the Physics Department - subMIT** **MIT, USA**
Talk on "Emulating the atomic nucleus." Jan. 30 2025
- **Fall Meeting of the APS Division of Nuclear Physics** **MIT, USA**
Talk on "Ab initio theory towards reliable neutrinoless double beta decay NMEs." Oct. 10 2024
- **Fall Meeting of the APS Division of Nuclear Physics** **MIT, USA, Virtual**
Talk on "Ab initio neutrinoless double-beta decay matrix elements." Oct. 13 2021
- **Winter Nuclear & Particle Physics Conference** **Virtual**
Talk on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Feb. 10th 2021
- **Progress in Ab Initio Techniques in Nuclear Physics** **TRIUMF, CA**
Talk and poster on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Mar. 5 2020
- **55th Canadian Undergraduate Physics Conference** **McGill University, CA**
Talk and poster on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Nov. 8 2019
- **McGill Physics Department Poster session** **McGill University, CA**
Poster on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Sept. 6 2019
- **TRIUMF User Group AGM Poster Slam! Session** **TRIUMF, CA**
Poster on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Aug. 22 2019
- **TRIUMF Student Symposium** **TRIUMF, CA**
Talk on "Ab initio Double-Beta Decay Nuclear Matrix Elements." Aug. 15 2019
- **McGill Undergraduate Research Conference** **McGill University, CA**
Poster on the "Development of an electroluminescent light source." Oct. 4 2018
- **McGill Physics Department Poster session** **McGill University, CA**
Poster on the "Development of an electroluminescent light source." Sept. 05 2018

Evaluation Committee Meetings.....

- **The NRC's Peer Review Committee quinquennial evaluation** **TRIUMF, CA**
Poster on "Towards reliable nuclear matrix elements for neutrinoless double beta decay". Nov. 29 2022
1/2 of the appointed TRIUMF theory department representative
- **Biannual Advisory Committee of TRUMF (ACOT) meeting** **TRIUMF, CA**
Poster on "Towards reliable nuclear matrix elements for neutrinoless double beta decay". Oct. 25 2022
1/2 of the appointed TRIUMF theory department representative
- **Biannual Advisory Committee of TRUMF (ACOT) meeting** **TRIUMF, CA**
Talk on "Ab initio neutrinoless double beta decay nuclear matrix elements". Apr. 26 2022
1/2 of the appointed TRIUMF theory department representative

Journal Articles

- [1] **A. Belley**, B. Romeo, J. Engel, D. Kekejian, T. Miyagi, S. Foster, P. Navratil, B. C. He, S. R. Stroberg, J. D. Holt, and R. F. G. Ruiz, "Ab initio calculation of symmetry-breaking observables," 2026, Submitted to Physical Review Letters. arXiv: 2605.11353 [nucl-th].
- [2] A. Brinson, B. Rickey, P. Arthuis, **A. Belley**, S. Campbell, X. Chen, A. Dockery, S. Elhatisari, H. Erington, N. Gamage, R. F. G. Ruiz, M. Heinz, C. Ireland, C. Izzo, C. Jones, J. Kartheim, K. König, D. Lee, Y.-Z. Ma, F. Maier, U.-G. Meißner, K. Minamisono, M. Moenter, J. Munoz, W. Nörtershäuser, A. Ortiz-Cortés,

- M. J. Palmes, S. Papa, F. P. Cruz, R. Ringle, H. Sims, C. Sumithrarachchi, A. Vernon, T. Wang, S. Wilkins, R. Yadav, and S. Zhang, "Nuclear charge radii of aluminium isotopes at the proton drip line," 2026, Submitted to Physical Review Letters. arXiv: 2605.09139 [nucl-ex].
- [3] D. Lange, D. Atanasov, M. Au, **A. Belley**, M. Benhatchi, K. Blaum, R. B. Cakirli, P. F. Giesel, A. Herlert, J. D. Holt, B. S. Hu, A. Jaries, C. Klink, Y. A. Litvinov, D. Lunney, V. Manea, F. Mehlhorn, T. Miyagi, M. Mougeot, S. Naimi, L. Nies, M. Schlaich, C. Schweiger, L. Schweikhard, T. Shickele, A. Todd, and W. Wojtaczka, "Constraining the trend of the $N = 50$ shell gap towards ^{100}Sn with the masses of $^{96-98}\text{Cd}$," 2026, Submitted to Physical Review Letters. arXiv: 2604.18287 [nucl-ex].
 - [4] J. M. Munoz, **A. Belley**, A. Ekström, G. Hagen, J. D. Holt, and R. F. G. Ruiz, "Linking electromagnetic moments to nuclear interactions with a global physics-driven machine-learning emulator," 2026, Submitted to Physical Review X. arXiv: 2603.26905 [nucl-th].
 - [5] S.-M. Udrescu, **A. Belley**, J. D. Holt, G. Perez, and R. F. G. Ruiz, "Probing the electroweak structure of nuclei with rare atoms and molecules," 2026, In preparation with Review of Modern Physics. arXiv: 2608.18219 [physics.atom-ph].
 - [6] A. Todd, T. Shickele, **A. Belley**, L. Jokiniemi, and J. D. Holt, "Ab initio short-range nuclear matrix elements for neutrinoless double-beta decay," *Phys. Rev. Lett.*, vol. 137, p. 092501, 9 Aug. 2026. doi: 10.1103/d2n7-n2c7.
 - [7] T. Shickele, L. Jokiniemi, **A. Belley**, and J. D. Holt, "Global ab initio neutrino mass limits from neutrinoless double-beta decay," *Phys. Rev. D*, vol. 114, p. 013007, 1 Jul. 2026. doi: 10.1103/rsw5-57fz.
 - [8] **A. Belley**, J. M. Munoz, and R. F. Garcia Ruiz, "Global framework for emulation of nuclear calculations," *Phys. Rev. Lett.*, vol. 136, p. 082501, 8 Feb. 2026. doi: 10.1103/mvc3-qdtc.
 - [9] **A. Belley**, J. Pitcher, T. Miyagi, S. R. Stroberg, and J. D. Holt, "Correlation of $0\nu\beta\beta$ decay nuclear matrix elements with nucleon-nucleon phase shifts," *Phys. Rev. C*, vol. 113, p. 014319, 1 Jan. 2026. doi: 10.1103/cg41-189t.
 - [10] **A. Belley**, T. Miyagi, R. Stroberg, and J. D. Holt, "Constraining neutrinoless double beta decay matrix elements from ab initio nuclear theory," *AIP Conf. Proc.*, vol. 3143, no. 1, p. 020002, Feb. 2025. doi: 10.1063/5.0249855.
 - [11] **A. Belley**, J. M. Yao, B. Bally, J. Pitcher, J. Engel, H. Hergert, J. D. Holt, T. Miyagi, T. R. Rodríguez, A. M. Romero, S. R. Stroberg, and X. Zhang, "Ab Initio Uncertainty Quantification of Neutrinoless Double-Beta Decay in ^{76}Ge ," *Phys. Rev. Lett.*, vol. 132, p. 182502, 18 Apr. 2024. doi: 10.1103/PhysRevLett.132.182502.
 - [12] **A. Belley**, T. Miyagi, S. R. Stroberg, and J. D. Holt, "Ab initio calculations of neutrinoless $\beta\beta$ decay refine neutrino mass limits," Jul. 2023, Under review in Nature. arXiv: 2307.15156 [nucl-th].
 - [13] J. M. Yao, I. Ginnett, **A. Belley**, T. Miyagi, R. Wirth, S. Bogner, J. Engel, H. Hergert, J. D. Holt, and S. R. Stroberg, "Ab initio studies of the double-Gamow-Teller transition and its correlation with neutrinoless double- β decay," *Phys. Rev. C*, vol. 106, p. 014315, 1 Jul. 2022. doi: 10.1103/PhysRevC.106.014315.
 - [14] **A. Belley**, C. G. Payne, S. R. Stroberg, T. Miyagi, and J. D. Holt, "Ab Initio Neutrinoless Double-Beta Decay Matrix Elements for ^{48}Ca , ^{76}Ge , and ^{82}Se ," *Phys. Rev. Lett.*, vol. 126, p. 042502, 4 Jan. 2021. doi: 10.1103/PhysRevLett.126.042502.
 - [15] J. M. Yao, **A. Belley**, R. Wirth, T. Miyagi, C. G. Payne, S. R. Stroberg, H. Hergert, and J. D. Holt, "Ab initio benchmarks of neutrinoless double- β decay in light nuclei with a chiral Hamiltonian," *Phys. Rev. C*, vol. 103, p. 014315, 1 Jan. 2021. doi: 10.1103/PhysRevC.103.014315.
 - [16] **A. Belley**, "Development on an electroluminescent light source to characterize SiPMs for the nEXO collaboration," *Journal of Undergraduate Science and Technology Research*, vol. 1, no. 1, 2019.